

Your 8-hour Chemistry rescue plan

Everything that can show up in the Kerala **Plus Two (Class 12) Chemistry SAY exam 2026** — every chapter's important topics explained in plain language, how each topic gets asked, and practice questions with full model answers written the way the examiner wants them. Built around the real **60-mark theory pattern**.

📖 16 chapters

⌚ 2 hrs + 15 min cool-off

✍️ 1 / 2 / 3 / 4-mark questions

✅ Pass mark: 18 / 60 (30%)

🚫 No negative marking

🕒 How to spend your 8 hours

1. **0:00–0:20** Read the exam-pattern + answer-writing tips at the bottom first. Know how marks are won.
2. **0:20–3:30 Organic chemistry block** (Ch 10–13) — highest weightage, most predictable. Learn name-reactions + distinguishing tests cold.
3. **3:30–5:00 Physical block** — Electrochemistry, Chemical Kinetics, Solutions, Solid State. Practise the numericals; the formulas repeat.
4. **5:00–6:30 Inorganic block** — Coordination Compounds (CFT!), d & f-block ($\text{KMnO}_4/\text{K}_2\text{Cr}_2\text{O}_7$), p-block, Metallurgy.
5. **6:30–7:30** Biomolecules, Polymers, Surface Chemistry, Chemistry in Everyday Life — short, scoring, easy to memorise.
6. **7:30–8:00** Run **Quiz me**, re-read every "How it's asked" box, sleep if you can.

🎯 If you only have time for the big scorers

1. **Aldehydes/Ketones/Acids** — Tollens, Fehling, Iodoform, Aldol, Cannizzaro.
2. **Coordination Compounds** — IUPAC naming, CFT splitting, magnetic moment.
3. **Electrochemistry** — Nernst equation + EMF numericals.
4. **Chemical Kinetics** — first-order numericals, half-life, Arrhenius.
5. **Alcohols/Phenols/Ethers** — phenol acidity, Reimer–Tiemann, Williamson.
6. **Amines** — basicity order, diazonium reactions, distinguishing $1^\circ/2^\circ/3^\circ$.
7. **d & f-block** — KMnO_4 , $\text{K}_2\text{Cr}_2\text{O}_7$, lanthanoid contraction.
8. **Solutions** — colligative properties + van't Hoff factor.

These eight chapters alone typically cover well over half the paper. Master them first.

💡 Easy marks if you memorise the unit-cell formulas and defect types. Almost always one numerical (density / atoms per unit cell).

IMPORTANT TOPICS (EXPLAINED)

Crystalline vs amorphous solids

Crystalline solids have a long-range, regular, repeating arrangement of particles, sharp melting points, definite geometry and are *anisotropic* (properties differ with direction).

Examples: NaCl, diamond, quartz.

Amorphous solids have only short-range order, melt over a range, have no definite shape and are *isotropic*. Examples: glass, rubber, plastic. They are called *pseudo-solids* or supercooled liquids.

Unit cell & types of cubic cells

The smallest repeating unit of a crystal lattice is the **unit cell**. Number of atoms per unit cell (Z): contributions — corner atom = $1/8$, face atom = $1/2$, edge = $1/4$, body-centre = 1.

- **Simple cubic (SC):** $Z = 1$, packing efficiency 52.4%, coordination number 6.
- **Body-centred cubic (BCC):** $Z = 2$, packing efficiency 68%, coordination number 8.
- **Face-centred cubic (FCC)/CCP:** $Z = 4$, packing efficiency 74%, coordination number 12.

Close packing, voids & radius ratio

HCP and CCP/FCC are the closest packings (74% filled). For N close-packed atoms there are **N octahedral voids** and **$2N$ tetrahedral voids**.

Ionic solids: which void the cation occupies depends on the **radius ratio** r^+/r^- (tetrahedral 0.225–0.414, octahedral 0.414–0.732, cubic 0.732–1).

Density of a unit cell

The most common numerical. Use the density formula and remember to convert edge length to cm.

Point defects

Stoichiometric defects: *Schottky* (equal cations + anions missing → density decreases, seen in NaCl, KCl) and *Frenkel* (ion shifts to interstitial site → density unchanged, seen in AgCl, ZnS).

Impurity & non-stoichiometric defects: metal excess (F-centres → colour) and metal deficiency.

Electrical & magnetic properties

Conductors, insulators, semiconductors (band theory). Doping gives **n-type** (group 15 impurity, extra electrons) and **p-type** (group 13 impurity, holes).

Magnetism: paramagnetic, diamagnetic, ferromagnetic, antiferromagnetic, ferrimagnetic.

MUST-KNOW FORMULAS & REACTIONS

Density $d = (Z \cdot M) / (a^3 \cdot N_A)$ – Z = atoms/cell, M = molar mass (g/mol), a = edge (cm), $N_A = 6.022 \times 10^{23}$

Octahedral voids = N ; Tetrahedral voids = $2N$ (N = number of close-packed particles)

Packing efficiency: SC 52.4% · BCC 68% · FCC/HCP 74%

Relation in BCC: $4r = \sqrt{3} \cdot a$; in FCC: $4r = \sqrt{2} \cdot a$; in SC: $2r = a$

HOW QUESTIONS COME FROM THIS CHAPTER

► 1-mark: 'Which defect lowers density?' (Schottky). 'Coordination number of FCC?' (12).

► 2-mark: define Schottky vs Frenkel with one example each.

► 3-mark: density numerical — given edge length and type, find atomic mass or density.

► 3-mark: explain n-type and p-type semiconductors with the doping element.

PRACTICE QUESTIONS WITH MODEL ANSWERS

3 marks

An element has a body-centred cubic structure with edge length 288 pm and density 7.2 g/cm³. Calculate the atomic mass.

MODEL ANSWER

Formula: $d = Z \cdot M / (a^3 \cdot N_A) \Rightarrow M = d \cdot a^3 \cdot N_A / Z$.

BCC $\Rightarrow Z = 2$. $a = 288 \text{ pm} = 2.88 \times 10^{-8} \text{ cm}$, so $a^3 = 2.39 \times 10^{-23} \text{ cm}^3$.

$M = (7.2 \times 2.39 \times 10^{-23} \times 6.022 \times 10^{23}) / 2 \approx \mathbf{51.8 \text{ g/mol}}$.

Mark tip: write the formula line, the Z value, and the $a \rightarrow \text{cm}$ conversion — each is a step mark.

2 marks

Distinguish between Schottky and Frenkel defects.

MODEL ANSWER

- **Schottky:** equal numbers of cations and anions are missing; density decreases; shown by ionic solids with similar-sized ions (NaCl, KCl, CsCl).
- **Frenkel:** a smaller ion (usually cation) leaves its site and occupies an interstitial site; density unchanged; shown by solids with large size difference (AgCl, ZnS, AgBr).

1 mark

How many atoms are present in one face-centred cubic unit cell?

MODEL ANSWER

4 atoms ($8 \text{ corners} \times 1/8 + 6 \text{ faces} \times 1/2 = 1 + 3 = 4$).

2 marks

Why are amorphous solids called pseudo-solids or supercooled liquids?

MODEL ANSWER

Because they have only short-range order and no sharp melting point — they soften over a range of temperature and tend to flow very slowly like extremely viscous liquids (e.g. old glass panes thicken at the bottom). Structurally they resemble liquids, so they are 'super-cooled liquids'.

Ch 2

Solutions

HIGH

💡 A reliable source of one full numerical (colligative property) plus a definition. Learn the four colligative formulas and van't Hoff factor.

IMPORTANT TOPICS (EXPLAINED)

Concentration terms

- **Molarity (M)** = mol solute / L solution (temperature-dependent).
- **Molality (m)** = mol solute / kg solvent (temperature-independent — preferred for colligative work).
- **Mole fraction (x)** = mol component / total mol.
- **ppm, mass %, volume %.**

Henry's law & Raoult's law

Henry's law: the solubility of a gas in a liquid is proportional to its partial pressure: $p = K_H \cdot x$. Higher $K_H \implies$ lower solubility.

Raoult's law: for a solution of two volatile liquids, the partial vapour pressure of each component is proportional to its mole fraction: $p_A = p_A^\circ \cdot x_A$. Total pressure = $p_A^\circ x_A + p_B^\circ x_B$.

Ideal & non-ideal solutions

Ideal: obey Raoult's law at all concentrations, $\Delta H_{\text{mix}} = 0$, $\Delta V_{\text{mix}} = 0$ (e.g. benzene + toluene).

Positive deviation: weaker A–B forces, higher vapour pressure (ethanol + acetone) $\rightarrow$ minimum boiling azeotrope.

Negative deviation: stronger A–B forces, lower vapour pressure (chloroform + acetone) $\rightarrow$ maximum boiling azeotrope.

Colligative properties

Properties that depend on the **number** of solute particles, not their nature:

1. Relative lowering of vapour pressure
2. Elevation of boiling point (ΔT_b)
3. Depression of freezing point (ΔT_f)
4. Osmotic pressure (π)

Used to find molar mass of solute.

van't Hoff factor (i) & abnormal molar mass

$i = (\text{observed colligative effect})/(\text{normal effect}) = (\text{no. of particles after dissociation/association})/(\text{particles before})$.

- Electrolytes that dissociate: $i > 1$ ($\text{NaCl} \approx 2$, $\text{BaCl}_2 \approx 3$).
- Solutes that associate (e.g. acetic acid in benzene $\rightarrow$ dimer): $i < 1$.

MUST-KNOW FORMULAS & REACTIONS

Relative lowering of VP: $(p^\circ - p)/p^\circ = x_{\text{solute}}$

Elevation of boiling point: $\Delta T_b = i \cdot K_b \cdot m$

Depression of freezing point: $\Delta T_f = i \cdot K_f \cdot m$

Osmotic pressure: $\pi = i \cdot C \cdot R \cdot T$ (C in mol/L, $R = 0.0821 \text{ L atm K}^{-1}\text{mol}^{-1}$)

Molar mass from depression: $M = (i \cdot K_f \cdot w_B \cdot 1000)/(\Delta T_f \cdot w_A)$

HOW QUESTIONS COME FROM THIS CHAPTER

- ▶ 2-mark: state Henry's law / Raoult's law with one application.
- ▶ 3-mark: numerical — calculate molar mass from ΔT_b , ΔT_f or osmotic pressure.
- ▶ 2-mark: positive vs negative deviation with one example each.
- ▶ 1-mark: which colligative property is used to measure very large molar masses? (osmotic pressure).

PRACTICE QUESTIONS WITH MODEL ANSWERS

3 marks

1.0 g of a non-electrolyte solute dissolved in 50 g of benzene lowered the freezing point by 0.40 K. K_f for benzene = $5.12 \text{ K kg mol}^{-1}$. Find the molar mass of the solute.

MODEL ANSWER

Formula: $M = (K_f \cdot w_B \cdot 1000)/(\Delta T_f \cdot w_A)$ ($i = 1$, non-electrolyte).

$M = (5.12 \times 1.0 \times 1000)/(0.40 \times 50) = 5120/20 = \mathbf{256 \text{ g/mol}}$.

2 marks Define colligative property. Name the four colligative properties.

MODEL ANSWER

A **colligative property** depends only on the *number* of solute particles in a solution and not on their chemical nature.

The four are: (1) relative lowering of vapour pressure, (2) elevation of boiling point, (3) depression of freezing point, (4) osmotic pressure.

2 marks Why does a solution of chloroform and acetone show negative deviation from Raoult's law?

MODEL ANSWER

Acetone and chloroform form **intermolecular hydrogen bonds** ($\text{Cl}_3\text{C}-\text{H}\cdots\text{O}=\text{C}$), so the A–B attractions are stronger than the A–A and B–B attractions in the pure liquids. This lowers the escaping tendency, so the vapour pressure is lower than predicted by Raoult's law → negative deviation (and a maximum-boiling azeotrope).

1 mark What is the value of van't Hoff factor for NaCl in water (assuming complete dissociation)?

MODEL ANSWER

$i = 2$ ($\text{NaCl} \rightarrow \text{Na}^+ + \text{Cl}^-$).

Ch 3 Electrochemistry

HIGH

💡 Top-priority physical chapter. Expect a Nernst-equation or EMF numerical plus conductance theory. The formulas repeat every year.

IMPORTANT TOPICS (EXPLAINED)

Galvanic (voltaic) cell & electrode potential

A galvanic cell converts chemical energy into electrical energy. **Oxidation at anode (–), reduction at cathode (+)**. Standard hydrogen electrode (SHE) is the reference, $E^\circ = 0 \text{ V}$.

Cell notation: anode | anode solution || cathode solution | cathode.

Cell EMF & Nernst equation

$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$ (both as reduction potentials). The Nernst equation gives EMF at non-standard conditions.

EMF, Gibbs energy & equilibrium constant

$\Delta G^\circ = -nFE^\circ_{\text{cell}}$ links electrochemistry to thermodynamics. A positive $E^\circ_{\text{cell}} \Rightarrow$ negative $\Delta G^\circ \Rightarrow$ spontaneous. Also E°_{cell} relates to the equilibrium constant K_c .

Conductance & Kohlrausch's law

Conductivity (κ) decreases with dilution; **molar conductivity** (Λ_m) increases with dilution. For strong electrolytes Λ_m vs $\sqrt{c}$ is linear (Debye–Hückel–Onsager). **Kohlrausch's law:** limiting molar conductivity is the sum of ionic contributions — used to find Λ°_m of weak electrolytes.

Electrolysis & Faraday's laws

First law: mass deposited $\propto$ charge ($w = Z \cdot I \cdot t$). **Second law:** for the same charge, masses are proportional to equivalent masses. 1 Faraday = 96500 C = charge of 1 mole of electrons.

Batteries, fuel cells & corrosion

Primary (dry cell, mercury) — not rechargeable. **Secondary** (lead storage, Ni–Cd) — rechargeable. **Fuel cell** (H_2 – O_2) — continuous, pollution-free. **Corrosion** of iron is an electrochemical process; prevented by galvanising, painting, sacrificial protection.

MUST-KNOW FORMULAS & REACTIONS

$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$$

$$\text{Nernst (298 K): } E_{\text{cell}} = E^\circ_{\text{cell}} - (0.0591/n) \cdot \log Q$$

$$\Delta G^\circ = -nFE^\circ_{\text{cell}} \quad (F = 96500 \text{ C})$$

$$\log K_c = (n \cdot E^\circ_{\text{cell}})/0.0591$$

Molar conductivity: $\Lambda_m = (\kappa \times 1000)/c$ (c in mol/L)

Kohlrausch: $\Lambda^\circ_m = \nu_+ \cdot \lambda^\circ_+ + \nu_- \cdot \lambda^\circ_-$

Faraday's first law: $w = (M \cdot I \cdot t)/(n \cdot F)$

HOW QUESTIONS COME FROM THIS CHAPTER

- ▶ 3-mark: calculate E_{cell} using the Nernst equation given concentrations.
- ▶ 3-mark: calculate Λ°_m of a weak electrolyte (e.g. acetic acid) using Kohlrausch's law.
- ▶ 2-mark: differences between primary and secondary cells / write electrode reactions of a given cell.
- ▶ 1-mark: value of 1 Faraday / which electrode is the anode in a galvanic cell.

PRACTICE QUESTIONS WITH MODEL ANSWERS

3 marks Calculate the EMF of the cell $\text{Zn}|\text{Zn}^{2+}(0.1 \text{ M})||\text{Cu}^{2+}(1.0 \text{ M})|\text{Cu}$ at 298 K. Given $E^\circ(\text{Zn}^{2+}/\text{Zn}) = -0.76 \text{ V}$, $E^\circ(\text{Cu}^{2+}/\text{Cu}) = +0.34 \text{ V}$.

MODEL ANSWER

$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}} = 0.34 - (-0.76) = \mathbf{1.10 \text{ V}}$.

$n = 2$. Reaction: $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$, $Q = [\text{Zn}^{2+}]/[\text{Cu}^{2+}] = 0.1/1.0 = 0.1$.

$E_{\text{cell}} = 1.10 - (0.0591/2) \cdot \log(0.1) = 1.10 - (0.02955)(-1) = 1.10 + 0.0296 = \mathbf{1.13 \text{ V}}$.

2 marks State Kohlrausch's law and give one application.

MODEL ANSWER

Kohlrausch's law: at infinite dilution, the limiting molar conductivity of an electrolyte is the sum of the individual limiting molar conductivities of its cations and anions.

Application: calculating Λ°_m of a weak electrolyte (like acetic acid) which cannot be found by extrapolation; also for finding the degree of dissociation ($\alpha = \Lambda_m/\Lambda^\circ_m$).

2 marks

Why does molar conductivity increase on dilution while conductivity decreases?**MODEL ANSWER**

On dilution the number of ions per unit volume falls, so **conductivity (κ)** — which measures conducting power per unit volume — decreases. But **molar conductivity** accounts for all ions from one mole of electrolyte; on dilution more of the electrolyte dissociates and ions move more freely, so Λ_m increases and reaches a maximum (Λ°_m) at infinite dilution.

1 mark

What is the value of one Faraday of charge?**MODEL ANSWER**

96500 C (charge carried by one mole of electrons).

Ch 4

Chemical Kinetics

HIGH

💡 Numerical-heavy and predictable. The first-order integrated equation and half-life appear almost every year. Learn order vs molecularity cold.

IMPORTANT TOPICS (EXPLAINED)**Rate of reaction & rate law**

Rate = change in concentration per unit time. **Rate law**: $\text{Rate} = k[A]^x[B]^y$, where x, y are found experimentally. **Order** = $x + y$ (sum of powers). **Molecularity** = number of species in an elementary step (always a whole number ≥ 1).

Order vs molecularity

- Order is experimental, can be 0, fractional, even negative; defined for overall reaction.
- Molecularity is theoretical, a positive integer, defined only for elementary steps.
- The slowest step is the **rate-determining step**.

Integrated rate equations

Zero order: $[R] = [R]_0 - kt$; units of $k = \text{mol L}^{-1} \text{s}^{-1}$.

First order: $k = (2.303/t) \cdot \log([R]_0/[R])$; units of $k = \text{s}^{-1}$.

Half-life

First order: $t_{1/2} = 0.693/k$ — independent of initial concentration. **Zero order:** $t_{1/2} = [R]_0/2k$.

Pseudo first order reactions

Reactions that are actually higher order but behave as first order because one reactant (often water) is in large excess and its concentration stays effectively constant. Example: acid hydrolysis of an ester, inversion of cane sugar.

Temperature dependence — Arrhenius equation & collision theory

$k = A \cdot e^{(-E_a/RT)}$. Higher temperature $\Rightarrow$ more molecules cross the activation-energy barrier $\Rightarrow$ faster rate (rate roughly doubles per 10 °C). **Activation energy (E_a)** is the minimum energy needed; a catalyst lowers E_a by providing an alternate path.

MUST-KNOW FORMULAS & REACTIONS

First order: $k = (2.303/t) \cdot \log([R]_0/[R])$

First-order half-life: $t_{1/2} = 0.693/k$

Zero order: $k = ([R]_0 - [R])/t$; $t_{1/2} = [R]_0/2k$

Arrhenius: $k = A \cdot e^{(-E_a/RT)}$

$\log(k_2/k_1) = (E_a/2.303R) \cdot (1/T_1 - 1/T_2)$

HOW QUESTIONS COME FROM THIS CHAPTER

▶ 3-mark: first-order numerical — find k , or time for a given fraction to react, or remaining concentration.

▶ 2-mark: difference between order and molecularity.

▶ 2-mark: explain pseudo first-order reaction with an example.

▶ 1-mark: units of rate constant for a first-order / zero-order reaction.

PRACTICE QUESTIONS WITH MODEL ANSWERS

3 marks

A first-order reaction has a rate constant of $1.15 \times 10^{-3} \text{ s}^{-1}$. How long will it take for 5 g of the reactant to reduce to 3 g?

MODEL ANSWER

Formula: $k = (2.303/t) \cdot \log([R]_0/[R]) \Rightarrow t = (2.303/k) \cdot \log([R]_0/[R])$.

$t = (2.303 / 1.15 \times 10^{-3}) \cdot \log(5/3) = 2002.6 \times \log(1.667) = 2002.6 \times 0.2219 \approx \mathbf{444 \text{ s}}$.

2 marks

Distinguish between order and molecularity of a reaction.

MODEL ANSWER

- **Order** is the sum of the powers of concentration terms in the experimentally-determined rate law; it can be zero, fractional or integral and applies to the overall reaction.
- **Molecularity** is the number of reacting species in an elementary reaction; it is always a whole number and has meaning only for elementary steps.

2 marks

What is a pseudo first-order reaction? Give one example.

MODEL ANSWER

A reaction that is not truly first order but follows first-order kinetics because one reactant is present in such large excess that its concentration remains practically constant. **Example:** acid-catalysed hydrolysis of ethyl acetate (water is in excess) — $\text{CH}_3\text{COOC}_2\text{H}_5 + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{COOH} + \text{C}_2\text{H}_5\text{OH}$.

1 mark

The half-life of a first-order reaction is independent of what?

MODEL ANSWER

The **initial concentration** of the reactant.

Ch 5

Surface Chemistry

CORE

💡 Short, descriptive and high-scoring. Adsorption types, colloids and Hardy–Schulze rule are easy 2–3 mark grabs.

IMPORTANT TOPICS (EXPLAINED)

Adsorption: physisorption vs chemisorption

Physisorption Chemisorption

- **Physisorption:** weak van der Waals forces, low enthalpy (20–40 kJ/mol), reversible, multilayer, decreases with temperature, not specific.
- **Chemisorption:** chemical bond, high enthalpy (80–240 kJ/mol), irreversible, monolayer, increases then decreases with temperature, highly specific.

Freundlich adsorption isotherm

$x/m = k \cdot p^{(1/n)}$ ($0 < 1/n < 1$). At high pressure adsorption becomes independent of pressure (saturation).

Catalysis

Homogeneous (catalyst same phase) and **heterogeneous** (different phase, e.g. contact process). **Enzyme catalysis** is highly specific and works under mild conditions. Catalysts lower activation energy without being consumed.

Colloids

Particle size 1–1000 nm. Classified by: physical state (sol, gel, emulsion, aerosol, foam), affinity to medium (**lyophilic** = solvent-loving, stable; **lyophobic** = solvent-hating, need stabiliser), and particle type (multimolecular, macromolecular, associated/micelles).

Properties of colloids

- **Tyndall effect** — scattering of light (path becomes visible).
- **Brownian motion** — zig-zag particle movement.
- **Electrophoresis** — movement of charged colloidal particles in an electric field.
- **Coagulation** — colloid settling on adding electrolyte.

Hardy–Schulze rule & emulsions

Hardy–Schulze rule: the greater the valency of the coagulating (oppositely charged) ion, the greater its coagulating power. For a negative sol, $\text{Al}^{3+} > \text{Ba}^{2+} > \text{Na}^{+}$.

Emulsions: liquid-in-liquid colloids (O/W like milk, W/O like butter), stabilised by emulsifiers (soap).

MUST-KNOW FORMULAS & REACTIONS

Freundlich isotherm: $x/m = k \cdot p^{(1/n)}$ ($1/n$ between 0 and 1)

Coagulating power order (Hardy–Schulze): higher valency of opposite ion $\Rightarrow$ stronger coagulant

HOW QUESTIONS COME FROM THIS CHAPTER

- ▶ 2-mark: physisorption vs chemisorption (give 2 differences).
- ▶ 2-mark: lyophilic vs lyophobic colloids / Tyndall effect explanation.
- ▶ 1-mark: state Hardy–Schulze rule / give an example of O/W emulsion.
- ▶ 3-mark: explain electrophoresis, coagulation, and Brownian motion.

PRACTICE QUESTIONS WITH MODEL ANSWERS

2 marks Give two differences between physisorption and chemisorption.

MODEL ANSWER

- Physisorption involves weak van der Waals forces with low enthalpy (20–40 kJ/mol); chemisorption involves chemical bonds with high enthalpy (80–240 kJ/mol).
- Physisorption is reversible and forms multilayers; chemisorption is irreversible and forms a monolayer. Physisorption decreases with rising temperature whereas chemisorption first increases then decreases.

2 marks

What is the Tyndall effect and why is it shown by colloids but not true solutions?

MODEL ANSWER

The **Tyndall effect** is the scattering of a beam of light by colloidal particles, making the path of light visible. Colloidal particles (1–1000 nm) are large enough to scatter light, while particles in a true solution are too small (< 1 nm) to scatter it, so true solutions show no Tyndall effect.

1 mark

State the Hardy–Schulze rule.

MODEL ANSWER

The coagulating power of an ion increases with its **valency**; the ion with charge opposite to the sol and higher valency is the most effective coagulant (e.g. $\text{Al}^{3+} > \text{Ba}^{2+} > \text{Na}^{+}$ for a negative sol).

Ch 6

General Principles & Processes of Isolation of Elements

CORE

💡 Mostly memory-based. Ellingham diagram reasoning and named processes (roasting, calcination, electrolytic reduction) are common.

IMPORTANT TOPICS (EXPLAINED)

Concentration of ore

Removing gangue: **hydraulic washing** (gravity), **magnetic separation**, **froth flotation** (sulphide ores, uses collectors/depressants), **leaching** (chemical, e.g. bauxite with NaOH, gold/silver with NaCN).

Conversion to oxide

- **Calcination:** heating in limited air to remove moisture/ CO_2 from carbonate or hydrated ores.
- **Roasting:** heating sulphide ore strongly in excess air to form oxide + SO_2 .

Reduction — Ellingham diagram

The Ellingham diagram plots ΔG° vs T for oxide formation. A metal/agent whose line is **lower** can reduce the oxide of a metal whose line is higher. Carbon (C/CO) becomes a good reducing agent at high temperature (its line slopes downward). This explains why coke reduces iron oxide in a blast furnace but not very reactive metals.

Electrolytic reduction

Highly reactive metals (Na, Al, Mg) are extracted by electrolysis of their molten salts/oxides.

Aluminium — Hall–Héroult process: electrolysis of Al_2O_3 dissolved in molten cryolite (Na_3AlF_6) which lowers the melting point and increases conductivity.

Refining

- **Electrolytic refining** (Cu, Ag) — impure metal anode, pure metal cathode.
- **Zone refining** (Si, Ge for semiconductors) — based on different solubility of impurities in molten vs solid.
- **Vapour phase** — Mond process (Ni), van Arkel (Ti, Zr).

MUST-KNOW FORMULAS & REACTIONS

Roasting (sulphide): $2\text{ZnS} + 3\text{O}_2 \rightarrow 2\text{ZnO} + 2\text{SO}_2$

Calcination (carbonate): $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$

Blast furnace: $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$

Hall–Héroult: $2\text{Al}_2\text{O}_3 \rightarrow 4\text{Al} + 3\text{O}_2$ (molten, cryolite)

HOW QUESTIONS COME FROM THIS CHAPTER

- ▶ 2-mark: difference between roasting and calcination with equations.
- ▶ 3-mark: role of cryolite / extraction of aluminium by Hall–Héroult process.
- ▶ 2-mark: principle of zone refining / froth flotation.
- ▶ 1-mark: which reducing agent is used in a blast furnace? (CO / coke).

PRACTICE QUESTIONS WITH MODEL ANSWERS

2 marks

Distinguish between roasting and calcination with one example each.

MODEL ANSWER

Roasting: the sulphide ore is heated strongly in the presence of excess air to convert it to the oxide.
e.g. $2\text{ZnS} + 3\text{O}_2 \rightarrow 2\text{ZnO} + 2\text{SO}_2$.

Calcination: the ore (carbonate/hydrated) is heated in limited or no air to drive off CO_2 or water, giving the oxide. e.g. $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$.

3 marks

Describe the role of cryolite in the electrolytic extraction of aluminium.

MODEL ANSWER

In the Hall–Héroult process, pure Al_2O_3 has a very high melting point ($\sim 2050^\circ\text{C}$). Dissolving it in molten **cryolite** (Na_3AlF_6) with some fluorspar: (i) lowers the melting point of the mixture to about 950°C , (ii) increases the electrical conductivity of the melt, and (iii) makes the Al_2O_3 readily ionise. Aluminium is then deposited at the cathode and O_2 liberated at the carbon anode (which slowly burns away as CO/CO_2).

1 mark

Name the method used to refine germanium for semiconductor use.

MODEL ANSWER

Zone refining.

Ch 7

The p-Block Elements (Groups 15–18)

MEDIUM

 Fact-dense but rewarding. Contact process, preparation of NH_3/HNO_3 , structures of oxoacids and Xe compounds recur.

IMPORTANT TOPICS (EXPLAINED)

Group 15 — Nitrogen family

General trend: oxidation states -3 to $+5$. **Dinitrogen (N_2)** is inert due to a strong triple bond. **Ammonia (NH_3)** — Haber process, pyramidal, basic. **Nitric acid (HNO_3)** — Ostwald process. **Oxides of nitrogen** (N_2O , NO , NO_2 ...). Phosphorus allotropes (white = reactive, red = stable). **PCl_3 , PCl_5** structures (trigonal pyramidal; trigonal bipyramidal). Oxoacids of P (H_3PO_3 dibasic, H_3PO_4 tribasic).

Group 16 — Oxygen family

Ozone (O_3) — bent, powerful oxidiser. **Sulphur dioxide (SO_2)**. **Sulphuric acid (H_2SO_4)** — **Contact process** ($2SO_2 + O_2 \rightleftharpoons 2SO_3$ over V_2O_5 ; SO_3 absorbed in H_2SO_4 to give oleum). Oxoacids of sulphur (H_2SO_4 , H_2SO_3 , $H_2S_2O_7$).

Group 17 — Halogens

Most reactive non-metals; oxidising power $F_2 > Cl_2 > Br_2 > I_2$. **Chlorine** — bleaching, prepared from $MnO_2 + HCl$. **HCl . Interhalogens** (ClF_3 , IF_7). Oxoacids of chlorine ($HOCl$, $HClO_2$, $HClO_3$, $HClO_4$ — acid strength increases with O). Anomalous behaviour of fluorine.

Group 18 — Noble gases

Very low reactivity (stable octet). **Xenon fluorides** XeF_2 (linear), XeF_4 (square planar), XeF_6 ; XeO_3 (pyramidal). Used to illustrate VSEPR and hybridisation.

MUST-KNOW FORMULAS & REACTIONS

Haber: $N_2 + 3H_2 \rightleftharpoons 2NH_3$ (Fe catalyst, $\sim 450^\circ C$, high P)

Ostwald (HNO_3): $4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$; $2NO + O_2 \rightarrow 2NO_2$; $3NO_2 + H_2O \rightarrow 2HNO_3 + NO$

Contact (H_2SO_4): $2SO_2 + O_2 \rightleftharpoons 2SO_3$ (V_2O_5) ; $SO_3 + H_2SO_4 \rightarrow H_2S_2O_7$ (oleum) ; $+ H_2O \rightarrow 2H_2SO_4$

XeF_2 linear (sp^3d), XeF_4 square planar (sp^3d^2), XeF_6 distorted octahedral (sp^3d^3)

HOW QUESTIONS COME FROM THIS CHAPTER

- ▶ 3-mark: manufacture of H_2SO_4 by the contact process / HNO_3 by the Ostwald process (with equations).
- ▶ 2-mark: why is N_2 inert? / why does fluorine show anomalous behaviour?
- ▶ 2-mark: structures and hybridisation of XeF_2 , XeF_4 , XeF_6 .
- ▶ 1-mark: shape of NH_3 / which is the strongest oxoacid of chlorine? (HClO_4).

PRACTICE QUESTIONS WITH MODEL ANSWERS

3 marks

Describe the manufacture of sulphuric acid by the Contact process.

MODEL ANSWER

Step 1: Sulphur or a sulphide is burnt to SO_2 : $\text{S} + \text{O}_2 \rightarrow \text{SO}_2$.

Step 2 (key step): SO_2 is oxidised to SO_3 over a V_2O_5 catalyst at $\sim 450^\circ\text{C}$: $2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3$ (exothermic, so moderate temperature and high pressure favour SO_3).

Step 3: SO_3 is absorbed in concentrated H_2SO_4 to form oleum: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$.

Step 4: Oleum is diluted with water: $\text{H}_2\text{S}_2\text{O}_7 + \text{H}_2\text{O} \rightarrow 2\text{H}_2\text{SO}_4$. (SO_3 is not absorbed directly in water because it forms a fog.)

2 marks

Why is dinitrogen (N_2) chemically inert at room temperature?

MODEL ANSWER

The two nitrogen atoms are held by a very strong **$\text{N}\equiv\text{N}$ triple bond** with a high bond dissociation enthalpy ($\sim 941 \text{ kJ/mol}$). A large amount of energy is needed to break this bond, so N_2 does not react easily under ordinary conditions.

2 marks

Give the shapes and hybridisation of XeF_2 and XeF_4 .

MODEL ANSWER

XeF_2 : linear shape, sp^3d hybridisation (3 lone pairs in equatorial positions).

XeF_4 : square planar shape, sp^3d^2 hybridisation (2 lone pairs above and below the plane).

1 mark What is the shape of the ammonia (NH₃) molecule?

MODEL ANSWER

Trigonal pyramidal (sp³ hybridised with one lone pair).

Ch 8 The d- and f-Block Elements

HIGH

💡 Reliable scorer. Properties of transition metals + the prep/properties of KMnO₄ and K₂Cr₂O₇ + lanthanoid contraction are near-certain.

IMPORTANT TOPICS (EXPLAINED)

General characteristics of transition elements

- Partly filled d-orbitals in the element or its ions.
- **Variable oxidation states** (because (n-1)d and ns energies are close).
- **Coloured ions** (d-d transitions).
- **Paramagnetism** (unpaired d electrons).
- Good **catalysts**, form **complexes**, **interstitial compounds**, and **alloys**.

Magnetic moment

Spin-only magnetic moment $\mu = \sqrt{n(n+2)}$ BM, where n = number of unpaired electrons. Used to find unpaired electrons / oxidation state.

Potassium permanganate (KMnO₄)

Prepared from pyrolusite (MnO₂): fuse with KOH + air/oxidant → K₂MnO₄ (green), then oxidise/electrolyse → KMnO₄ (purple). Strong oxidising agent in acidic medium (Mn⁷⁺ → Mn²⁺, gains 5 e⁻).

Potassium dichromate (K₂Cr₂O₇)

From chromite ore; oxidising agent in acidic medium (Cr⁶⁺ → Cr³⁺, gains 3 e⁻ per Cr, 6 per dichromate). Orange in acid, converts to yellow chromate in base (interconversion equilibrium).

Lanthanoids & lanthanoid contraction

Lanthanoid contraction: the steady decrease in atomic/ionic radii across the 4f series because the poor shielding by 4f electrons lets the increasing nuclear charge pull the shells inward. **Consequences:** similarity of 4d and 5d elements (Zr–Hf), difficulty in separating lanthanoids, gradual change in basicity.

Actinoids

5f series, all radioactive, show a wider range of oxidation states than lanthanoids; actinoid contraction is greater due to even poorer 5f shielding.

MUST-KNOW FORMULAS & REACTIONS

Spin-only magnetic moment: $\mu = \sqrt{n(n+2)}$ BM (n = unpaired electrons)

KMnO₄ in acid: $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$

K₂Cr₂O₇ in acid: $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$

Chromate–dichromate: $2\text{CrO}_4^{2-} + 2\text{H}^+ \rightleftharpoons \text{Cr}_2\text{O}_7^{2-} + \text{H}_2\text{O}$

HOW QUESTIONS COME FROM THIS CHAPTER

- ▶ 3-mark: preparation and oxidising action of KMnO₄ or K₂Cr₂O₇ (with half-reaction).
- ▶ 2-mark: what is lanthanoid contraction and its consequences?
- ▶ 2-mark: calculate spin-only magnetic moment for a given ion (e.g. Fe²⁺, $n=4 \Rightarrow 4.9$ BM).
- ▶ 1-mark: why are transition-metal ions coloured? (d–d transition).

PRACTICE QUESTIONS WITH MODEL ANSWERS

3 marks

What is lanthanoid contraction? Mention two of its consequences.**MODEL ANSWER**

Lanthanoid contraction is the gradual decrease in the atomic and ionic radii of the lanthanoids with increasing atomic number, caused by the poor shielding effect of the 4f electrons, which allows the increasing nuclear charge to draw the electron shells closer.

Consequences: (1) The second and third transition series elements of the same group have almost identical sizes (e.g. Zr and Hf), making them hard to separate. (2) There is a small but steady decrease in the basic strength of the lanthanoid hydroxides across the series.

2 marks

Calculate the spin-only magnetic moment of the Fe^{2+} ion.**MODEL ANSWER**

$\text{Fe}^{2+} = [\text{Ar}]3d^6 \Rightarrow$ unpaired electrons $n = 4$.

$\mu = \sqrt{n(n+2)} = \sqrt{4 \times 6} = \sqrt{24} \approx \mathbf{4.90 \text{ BM}}$.

2 marks

Why do transition metals show variable oxidation states?**MODEL ANSWER**

Because the energies of the $(n-1)d$ and ns orbitals are very close, electrons from both can take part in bonding. So a transition metal can lose different numbers of electrons and exhibit several oxidation states differing by one (e.g. $\text{Fe}^{2+}/\text{Fe}^{3+}$, Mn^{2+} to Mn^{7+}).

1 mark

Why are most transition-metal compounds coloured?**MODEL ANSWER**

Because of **d–d electronic transitions**: an electron absorbs visible light to jump between split d-orbitals, and the complementary colour is transmitted.

Ch 9 Coordination Compounds

HIGH

💡 One of the highest-yield chapters. IUPAC naming, CFT splitting/CFSE, magnetic behaviour and isomerism are exam favourites.

IMPORTANT TOPICS (EXPLAINED)

Werner's theory & basic terms

Ligand — donor species (Lewis base). **Coordination number** — number of donor atoms bonded to the central metal. **Denticity**: monodentate (Cl^- , NH_3), bidentate (en, ox), polydentate (EDTA, hexadentate). **Chelate** — ring formed by a polydentate ligand (extra stability). **Ambidentate** ligand can bind through either of two atoms ($-\text{NO}_2$ vs $-\text{ONO}$; $-\text{SCN}$ vs $-\text{NCS}$).

IUPAC nomenclature

Rules: name ligands alphabetically before the metal; anionic ligands end in -o (chlorido, cyanido); use di/tri or bis/tris for multiples; cation named first then anion; oxidation state of metal in Roman numerals; complex anion ends in -ate.

Isomerism

Structural: ionisation, linkage, coordination, hydrate. **Stereoisomerism**: geometrical (cis-trans), optical (mirror images).

Valence Bond Theory (VBT)

Explains shape & magnetism via hybridisation: octahedral d^2sp^3 (inner, low-spin) or sp^3d^2 (outer, high-spin); tetrahedral sp^3 ; square planar dsp^2 .

Crystal Field Theory (CFT)

Ligands split the d-orbitals. **Octahedral**: t_{2g} (lower) and e_g (upper), splitting Δ_o . **Tetrahedral**: reversed, $\Delta_t = (4/9)\Delta_o$ (always high-spin). **Spectrochemical series** ranks ligand field strength: $\text{I}^- < \text{Br}^- < \text{Cl}^- < \text{F}^- < \text{OH}^- < \text{H}_2\text{O} < \text{NH}_3 < \text{en} < \text{CN}^- \approx \text{CO}$. Strong-field ligands $\Rightarrow$ large $\Delta \Rightarrow$ pairing $\Rightarrow$ low-spin; weak-field $\Rightarrow$ high-spin. Colour and magnetism follow from the splitting.

Stability & applications

Chelate effect increases stability. Applications: EDTA in water-hardness titration, haemoglobin (Fe), chlorophyll (Mg), vitamin B_{12} (Co), extraction of metals (cyanide process), electroplating.

MUST-KNOW FORMULAS & REACTIONS

CFSE (octahedral) = $(-0.4 \times n_{t_2g} + 0.6 \times n_{e_g}) \Delta_0$ (+ pairing energy terms)

$\Delta_t = (4/9) \Delta_0$ – tetrahedral splitting always smaller $\Rightarrow$ high spin

Magnetic moment $\mu = \sqrt{n(n+2)}$ BM (n = unpaired electrons in the complex)

Spectrochemical (weak-strong): I^-

HOW QUESTIONS COME FROM THIS CHAPTER

▶ 3-mark: write IUPAC name OR formula of given complexes; predict shape/magnetism using CFT.

▶ 3-mark: explain why $[\text{Fe}(\text{CN})_6]^{4-}$ is low-spin/diamagnetic but $[\text{FeF}_6]^{3-}$ is high-spin (spectrochemical series).

▶ 2-mark: define ligand, coordination number, chelate, ambidentate ligand.

▶ 2-mark: cis-trans / optical isomerism with an example.

▶ 1-mark: name the donor atom in NH_3 as a ligand / which is a stronger field ligand, CN^- or H_2O ?

PRACTICE QUESTIONS WITH MODEL ANSWERS

3 marks

Using crystal field theory, explain why $[\text{Co}(\text{NH}_3)_6]^{3+}$ is diamagnetic while $[\text{CoF}_6]^{3-}$ is paramagnetic.

MODEL ANSWER

Both contain $\text{Co}^{3+} = d^6$. The geometry is octahedral, so the d-orbitals split into t_{2g} and e_g with gap Δ_0 .

$[\text{Co}(\text{NH}_3)_6]^{3+}$: NH_3 is a **strong-field** ligand (high in the spectrochemical series), so Δ_0 is large — larger than the pairing energy. All six electrons pair up in the t_{2g} set ($t_{2g}^6 e_g^0$). No unpaired electrons $\Rightarrow$ **diamagnetic**, low-spin.

$[\text{CoF}_6]^{3-}$: F^- is a **weak-field** ligand, so Δ_0 is small. Electrons fill all five d-orbitals before pairing ($t_{2g}^4 e_g^2$), leaving 4 unpaired electrons $\Rightarrow$ **paramagnetic**, high-spin.

3 marks Write the IUPAC names of: (a) $\text{K}_4[\text{Fe}(\text{CN})_6]$ (b) $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$ (c) $[\text{Co}(\text{en})_3]\text{Cl}_3$.

MODEL ANSWER

(a) Potassium hexacyanidoferrate(II)

(b) Hexaamminechromium(III) chloride

(c) Tris(ethane-1,2-diamine)cobalt(III) chloride

Tip: ligands alphabetical, anionic ligands end -o, complex anion ends -ate, oxidation state in Roman numerals.

2 marks Define a chelate and an ambidentate ligand with one example each.

MODEL ANSWER

Chelate: a ring structure formed when a polydentate ligand binds to a metal through two or more donor atoms, e.g. ethylenediamine (en) forming a five-membered ring. It gives extra stability (chelate effect).

Ambidentate ligand: a monodentate ligand that can coordinate through either of two different donor atoms, e.g. NO_2^- (through N as nitrito-N or O as nitrito-O) and SCN^- (through S or N).

1 mark Which is the stronger field ligand: H_2O or CN^- ?

MODEL ANSWER

CN^- (it is much higher in the spectrochemical series).

Ch 10 Haloalkanes and Haloarenes

HIGH

💡 Mechanisms are gold here. SN_1 vs SN_2 , optical inversion, and the low reactivity of haloarenes are repeat questions.

IMPORTANT TOPICS (EXPLAINED)

Nomenclature & classification

Classify as $1^\circ/2^\circ/3^\circ$ and as alkyl/allyl/benzyl/vinyl/aryl halides. Haloalkanes have $\text{C}(\text{sp}^3)\text{-X}$; haloarenes have $\text{C}(\text{sp}^2)\text{-X}$ attached to an aromatic ring.

Preparation

From alcohols (with HX, PCl_3 , PCl_5 , SOCl_2 — SOCl_2 is best, by-products are gases), from alkenes (HX addition, Markovnikov; or halogenation), and Finkelstein/Swartz reactions (halide exchange).

Nucleophilic substitution — $\text{S}_\text{N}1$ vs $\text{S}_\text{N}2$

$\text{S}_\text{N}2$: one step, bimolecular, $\text{rate} \propto [\text{substrate}][\text{Nu}]$; back-side attack $\Rightarrow$ **inversion of configuration (Walden inversion)**; favoured by primary halides + strong nucleophile + polar aprotic solvent.

$\text{S}_\text{N}1$: two steps via carbocation, unimolecular, $\text{rate} \propto [\text{substrate}]$; gives **racemisation**; favoured by tertiary halides + polar protic solvent. Reactivity $\text{S}_\text{N}1$: $3^\circ > 2^\circ > 1^\circ$.

Elimination (β -elimination)

Dehydrohalogenation with alcoholic KOH gives alkenes; **Saytzeff rule** — the more substituted (more stable) alkene predominates.

Reactivity of haloarenes

Aryl halides are much **less reactive** towards nucleophilic substitution because of (i) resonance/partial double-bond character of C-X , (ii) sp^2 carbon (more electronegative, shorter stronger bond), and (iii) repulsion between the nucleophile and the electron-rich ring.

Polyhalogen compounds

Uses and concerns: **chloroform** (CHCl_3 , forms poisonous phosgene on air-oxidation), **CCl_4** , **DDT** (persistent insecticide), **freons** (CFCs — ozone depletion).

MUST-KNOW FORMULAS & REACTIONS

Alcohol $\rightarrow$ halide (best): $\text{R-OH} + \text{SOCl}_2 \rightarrow \text{R-Cl} + \text{SO}_2 + \text{HCl}$

$\text{S}_\text{N}2$: $\text{rate} = k[\text{R-X}][\text{Nu}]$, inversion; $\text{S}_\text{N}1$: $\text{rate} = k[\text{R-X}]$, racemisation

Reactivity SN1: $3^\circ > 2^\circ > 1^\circ$; SN2: $1^\circ > 2^\circ > 3^\circ$

Dehydrohalogenation: $\text{R-CH}_2\text{-CHX-R}' + \text{alc.KOH} \rightarrow \text{alkene (Saytzeff)}$

HOW QUESTIONS COME FROM THIS CHAPTER

- ▶ 3-mark: distinguish SN1 and SN2 mechanisms (with rate law and stereochemistry).
- ▶ 2-mark: why are haloarenes less reactive than haloalkanes towards nucleophilic substitution?
- ▶ 2-mark: what is Walden inversion / Saytzeff rule?
- ▶ 1-mark: best reagent to convert an alcohol to a chloroalkane (SOCl_2).

PRACTICE QUESTIONS WITH MODEL ANSWERS

3 marks

Compare SN1 and SN2 reaction mechanisms with respect to molecularity, kinetics, stereochemistry and substrate preference.

MODEL ANSWER

SN1 SN2

- **Steps/molecularity:** SN1 = two steps, unimolecular (carbocation intermediate); SN2 = one step, bimolecular.
- **Rate law:** SN1 rate = $k[\text{R-X}]$; SN2 rate = $k[\text{R-X}][\text{Nu}]$.
- **Stereochemistry:** SN1 gives racemisation; SN2 gives inversion (Walden inversion).
- **Substrate:** SN1 favoured by 3° halides (stable carbocation); SN2 favoured by 1° halides (less steric hindrance).

2 marks

Why is chlorobenzene less reactive than chloromethane towards nucleophilic substitution?

MODEL ANSWER

In chlorobenzene the C–Cl bond has **partial double-bond character** due to resonance of the chlorine lone pair into the ring, making it shorter and stronger. The carbon is also **sp² hybridised** (more electronegative, holds the bond tightly), and the electron-rich ring repels the incoming nucleophile. All three factors make substitution difficult, unlike in chloromethane.

2 marks What is the Saytzeff rule? Illustrate with the dehydrohalogenation of 2-bromobutane.

MODEL ANSWER

Saytzeff rule: in an elimination reaction, the alkene that is more highly substituted (and hence more stable) is formed as the major product. From 2-bromobutane with alcoholic KOH, **but-2-ene** (more substituted) is the major product over but-1-ene.

1 mark Which reagent best converts ethanol to chloroethane with the cleanest by-products?

MODEL ANSWER

Thionyl chloride (SOCl_2) — the by-products SO_2 and HCl are gases that escape, giving pure product.

Ch 11 Alcohols, Phenols and Ethers

HIGH

💡 Heavy organic chapter. Acidity of phenol, Reimer–Tiemann, Kolbe, Williamson synthesis and distinguishing tests are near-certain.

IMPORTANT TOPICS (EXPLAINED)

Preparation of alcohols & phenols

Alcohols: hydration of alkenes (Markovnikov / hydroboration-anti-Markovnikov), reduction of carbonyls, Grignard reagent + carbonyl. **Phenols:** from cumene (cumene $\rightarrow$ phenol + acetone), from chlorobenzene (Dow process), from diazonium salts.

Acidity of phenol vs alcohol

Phenol is far more acidic than alcohols because the **phenoxide ion is resonance-stabilised** (negative charge delocalised into the ring). Electron-withdrawing groups ($-\text{NO}_2$, especially ortho/para) increase acidity; electron-donating groups decrease it.

Reactions of alcohols

Esterification (with acids), dehydration (to alkenes, conc. H_2SO_4), oxidation ($1^\circ \rightarrow$ aldehyde $\rightarrow$ acid; $2^\circ \rightarrow$ ketone; 3° resists), reaction with Na (H_2 evolved), Lucas test (distinguishes $1^\circ/2^\circ/3^\circ$).

Reactions of phenols

Electrophilic substitution is easy (-OH is activating, o/p-directing): bromination, nitration.

Kolbe's reaction (phenol $\rightarrow$ salicylic acid via CO_2). **Reimer–Tiemann reaction** (phenol + $\text{CHCl}_3 + \text{KOH} \rightarrow$ salicylaldehyde). Reaction with FeCl_3 gives a violet colour (test for phenol).

Ethers — Williamson synthesis

Williamson ether synthesis: alkoxide + primary alkyl halide $\rightarrow$ ether ($\text{S}_{\text{N}}2$; use primary halide to avoid elimination). Ethers undergo C–O cleavage with HI/HBr .

MUST-KNOW FORMULAS & REACTIONS

Williamson: $\text{R-O-Na}^+ + \text{R}'\text{-X} \rightarrow \text{R-O-R}' + \text{NaX}$ (use 1° $\text{R}'\text{X}$)

Kolbe: sodium phenoxide + CO_2 (high P) $\rightarrow$ sodium salicylate $\rightarrow$ salicylic acid

Reimer–Tiemann: phenol + CHCl_3 + aq. $\text{KOH} \rightarrow$ salicylaldehyde (-CHO at ortho)

Oxidation: 1° alcohol $\rightarrow$ aldehyde $\rightarrow$ carboxylic acid; 2° alcohol $\rightarrow$ ketone

HOW QUESTIONS COME FROM THIS CHAPTER

- ▶ 3-mark: explain why phenol is more acidic than ethanol; effect of substituents.
- ▶ 2-mark: Reimer–Tiemann / Kolbe reaction with equation.
- ▶ 2-mark: Williamson synthesis — why use a primary halide?
- ▶ 2-mark: distinguish 1° , 2° , 3° alcohols (Lucas test) / test for phenol (FeCl_3).
- ▶ 1-mark: product of oxidation of a secondary alcohol (ketone).

PRACTICE QUESTIONS WITH MODEL ANSWERS

3 marks Explain why phenol is more acidic than ethanol. How do nitro groups affect the acidity of phenol?

MODEL ANSWER

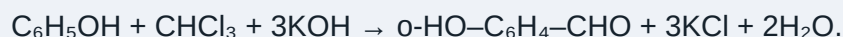
On losing a proton, phenol gives the **phenoxide ion** in which the negative charge is delocalised over the ring by resonance, making it stable; ethanol gives the ethoxide ion where the charge is localised on oxygen and is destabilised by the electron-donating ethyl group. Greater anion stability $\Rightarrow$ stronger acid, so phenol ($\text{pK}_a \approx 10$) is far more acidic than ethanol ($\text{pK}_a \approx 16$).

Nitro groups are electron-withdrawing; at the ortho/para positions they further delocalise the negative charge of the phenoxide ion, stabilising it. So nitrophenols are more acidic, and 2,4,6-trinitrophenol (picric acid) is a strong acid.

2 marks Write the Reimer–Tiemann reaction with equation.

MODEL ANSWER

When phenol is treated with chloroform (CHCl_3) and aqueous KOH at $\sim 340\text{ K}$, a $-\text{CHO}$ group is introduced at the **ortho** position, giving **salicylaldehyde** (2-hydroxybenzaldehyde) after acid hydrolysis.



2 marks Why is a primary alkyl halide preferred in Williamson's ether synthesis?

MODEL ANSWER

Williamson synthesis proceeds by an **$\text{S}_{\text{N}}2$** attack of the alkoxide on the alkyl halide. Alkoxides are strong bases as well as nucleophiles. With secondary or tertiary halides, the base promotes **elimination** (forming an alkene) instead of substitution. A primary halide has little steric hindrance, so substitution dominates and the ether forms in good yield.

1 mark What colour does phenol give with neutral FeCl_3 ?

MODEL ANSWER

A **violet/purple** colour (test for phenol).

Ch 12 Aldehydes, Ketones and Carboxylic Acids

HIGH

💡 Usually the single highest-weightage chapter. Named reactions and distinguishing tests (Tollens, Fehling, Iodoform) are guaranteed marks.

IMPORTANT TOPICS (EXPLAINED)

Nucleophilic addition

The carbonyl carbon (δ^+) is attacked by nucleophiles. **Reactivity:** aldehydes > ketones (less steric hindrance, less +I from one R group). Addition of HCN (cyanohydrin), NaHSO_3 , alcohols (acetal), ammonia derivatives (imines, oximes, hydrazones).

Aldol condensation

Carbonyl compounds with α -hydrogen, in dilute base, give a β -hydroxy carbonyl (aldol) which can dehydrate to an α,β -unsaturated carbonyl. Cross-aldol uses two different partners.

Cannizzaro reaction

Aldehydes with **no α -hydrogen** (e.g. HCHO, benzaldehyde) undergo disproportionation in concentrated base: one molecule is oxidised to the acid (salt) and another reduced to the alcohol.

Reduction reactions

Clemmensen (Zn-Hg/HCl) and **Wolff-Kishner** ($\text{NH}_2\text{NH}_2/\text{KOH}$) both reduce C=O to CH_2 . $\text{LiAlH}_4/\text{NaBH}_4$ reduce carbonyls to alcohols.

Distinguishing tests

- **Tollens' reagent** (ammoniacal AgNO_3): aldehyde gives a silver mirror; ketone does not.
- **Fehling's solution**: aliphatic aldehyde gives a red-brown Cu_2O precipitate; ketone does not.
- **Iodoform test** (I_2/NaOH): positive for ethanal and methyl ketones ($\text{CH}_3\text{CO-}$ group) and ethanol — yellow CHI_3 precipitate.

Carboxylic acids

Acidity is due to **resonance-stabilised carboxylate ion**; electron-withdrawing groups (Cl) increase acidity (e.g. trichloroacetic > acetic). Reactions: salt formation, esterification, conversion to acid chlorides/anhydrides/amides. **HVZ reaction** (α -halogenation with X_2 /red P).

MUST-KNOW FORMULAS & REACTIONS

Tollens': $RCHO + 2[Ag(NH_3)_2]^+ + 3OH^- \rightarrow RCOO^- + 2Ag\downarrow \text{ (mirror)} + \dots$

Iodoform: $CH_3CO-R \text{ (or } CH_3CH(OH)R) + 3I_2 + 4NaOH \rightarrow CHI_3\downarrow + RCOONa + \dots$

Cannizzaro: $2HCHO + NaOH \rightarrow CH_3OH + HCOONa \text{ (no } \alpha\text{-H)}$

Aldol: $2CH_3CHO \xrightarrow{\text{(dil. NaOH)}} CH_3CH(OH)CH_2CHO \rightarrow CH_3CH=CHCHO + H_2O$

Acidity: $Cl_3C-COOH > Cl_2CH-COOH > ClCH_2-COOH > CH_3COOH$

HOW QUESTIONS COME FROM THIS CHAPTER

- ▶ 3-mark: distinguish given pairs by chemical tests (e.g. ethanal vs propanone, acetophenone vs benzophenone).
- ▶ 3-mark: write/explain Aldol, Cannizzaro, Clemmensen, Wolff–Kishner with equations.
- ▶ 2-mark: why are aldehydes more reactive than ketones towards nucleophilic addition?
- ▶ 2-mark: which compounds give a positive iodoform test?
- ▶ 1-mark: reagent for the silver-mirror test (Tollens').

PRACTICE QUESTIONS WITH MODEL ANSWERS

3 marks

How would you distinguish between (a) ethanal and propanone, and (b) propanal and propanone using chemical tests?

MODEL ANSWER

(a) **Ethanal (CH_3CHO) vs propanone (CH_3COCH_3):** both give a positive iodoform test, so use **Tollens' reagent** — ethanal (an aldehyde) gives a silver mirror, propanone (a ketone) does not.

(b) **Propanal vs propanone:** use **Fehling's solution** — propanal (aldehyde) gives a red-brown precipitate of Cu_2O ; propanone gives no reaction. Alternatively Tollens' (propanal → silver mirror).

3 marks

What is the Cannizzaro reaction? Write it for benzaldehyde and explain why benzaldehyde undergoes it but acetaldehyde does not.

MODEL ANSWER

The **Cannizzaro reaction** is the base-induced disproportionation of an aldehyde that has **no α -hydrogen**: one molecule is oxidised to the carboxylate and another reduced to the alcohol.

$2\text{C}_6\text{H}_5\text{CHO} + \text{conc. NaOH} \rightarrow \text{C}_6\text{H}_5\text{COONa} + \text{C}_6\text{H}_5\text{CH}_2\text{OH}$ (sodium benzoate + benzyl alcohol).

Benzaldehyde has no α -hydrogen, so it cannot undergo aldol condensation and instead disproportionates. Acetaldehyde has α -hydrogens, so it prefers **aldol condensation** and does not give the Cannizzaro reaction.

2 marks

Why are aldehydes more reactive than ketones towards nucleophilic addition?

MODEL ANSWER

(1) **Electronic:** a ketone has two electron-donating alkyl/aryl groups that reduce the positive charge on the carbonyl carbon, whereas an aldehyde has only one, so the aldehyde carbon is more electrophilic.

(2) **Steric:** the two bulky groups in a ketone hinder the approach of the nucleophile more than the one group in an aldehyde. Both effects make aldehydes react faster.

1 mark

Which reagent gives a silver mirror with aldehydes?

MODEL ANSWER

Tollens' reagent (ammoniacal silver nitrate).

💡 Predictable scorer. Basicity order, preparation (Hoffmann, Gabriel), and diazonium-salt reactions appear most years.

IMPORTANT TOPICS (EXPLAINED)

Classification & preparation

1°, 2°, 3° amines and quaternary salts. Preparation: reduction of nitro compounds/nitriles/amides, ammonolysis of alkyl halides, **Gabriel phthalimide synthesis** (pure 1° amines only), **Hoffmann bromamide degradation** (amide → 1° amine with one fewer carbon).

Basicity of amines

Amines are basic (lone pair on N). In the gas phase basicity follows electron density, but in **aqueous solution** the order is governed by inductive effect + steric + solvation: typically 2° > 1° > 3° for many simple amines. **Aromatic amines (aniline) are weaker bases** than aliphatic amines because the N lone pair is delocalised into the ring.

Reactions of amines

Salt formation, acylation, reaction with nitrous acid (1° aliphatic → alcohol + N₂; 1° aromatic → diazonium salt), **carbylamine test** (1° amine + CHCl₃ + KOH → foul-smelling isocyanide — test for 1° amines), **Hinsberg's test** (distinguishes 1°/2°/3°).

Diazonium salts

Aryl diazonium salts (Ar-N₂⁺) are versatile intermediates. **Sandmeyer/Gattermann** (replace N₂⁺ by Cl, Br, CN), replacement by OH, H, I, F; **coupling reactions** with phenols/amines give azo dyes. They are key to making substituted aromatics that are hard to get directly.

MUST-KNOW FORMULAS & REACTIONS

Hoffmann bromamide: $\text{R-CONH}_2 + \text{Br}_2 + 4\text{NaOH} \rightarrow \text{R-NH}_2 + \text{Na}_2\text{CO}_3 + 2\text{NaBr} + 2\text{H}_2\text{O}$ (one C lost)

Gabriel: phthalimide $\rightarrow$ (KOH, R-X) $\rightarrow$ N-alkyl phthalimide $\rightarrow$ (hydrolysis) $\rightarrow$ R-NH₂ (pure 1°)

Diazotisation: Ar-NH₂ + NaNO₂ + 2HCl $\rightarrow$ Ar-N₂⁺Cl⁻ + NaCl + 2H₂O (0–5 °C)

Carbylamine (1° amine test): R-NH₂ + CHCl₃ + 3KOH $\rightarrow$ R-NC + 3KCl + 3H₂O

HOW QUESTIONS COME FROM THIS CHAPTER

- ▶ 3-mark: arrange amines in order of basic strength and justify (aniline vs methylamine etc.).
- ▶ 3-mark: synthetic uses of diazonium salts (Sandmeyer, coupling) with equations.
- ▶ 2-mark: Gabriel synthesis / Hoffmann bromamide degradation.
- ▶ 2-mark: distinguish 1°, 2°, 3° amines (Hinsberg / carbylamine).
- ▶ 1-mark: why can't aromatic amines be prepared by the Gabriel synthesis? (aryl halides don't undergo SN2).

PRACTICE QUESTIONS WITH MODEL ANSWERS

3 marks

Arrange the following in increasing order of basic strength in aqueous solution and justify: aniline, methylamine, ammonia.

MODEL ANSWER

Order: aniline < ammonia < methylamine.

Aniline is the weakest: the nitrogen lone pair is delocalised into the benzene ring by resonance, so it is less available for donation. **Methylamine** is the strongest: the electron-donating (+I) methyl group increases electron density on nitrogen, making the lone pair more available and the conjugate acid more stable. Ammonia lies in between.

3 marks Give the synthetic importance of diazonium salts with two example reactions.

MODEL ANSWER

Aryl diazonium salts let us replace the $-\text{N}_2^+$ group with many groups that are hard to introduce directly, giving a wide range of substituted aromatics.

Sandmeyer reaction: $\text{Ar}-\text{N}_2^+\text{Cl}^- + \text{CuCl} \rightarrow \text{Ar}-\text{Cl} + \text{N}_2$ (also gives $\text{Ar}-\text{Br}$, $\text{Ar}-\text{CN}$ with the right Cu salt).

Coupling reaction: $\text{Ar}-\text{N}_2^+ + \text{phenol} \rightarrow \text{p-hydroxyazobenzene}$ (an azo dye). These reactions are central to the manufacture of dyes.

2 marks Describe the carbylamine test and state what it identifies.

MODEL ANSWER

When a **primary amine** is heated with chloroform and alcoholic KOH, it forms an **isocyanide (carbylamine)** with an extremely offensive smell: $\text{R}-\text{NH}_2 + \text{CHCl}_3 + 3\text{KOH} \rightarrow \text{R}-\text{NC} + 3\text{KCl} + 3\text{H}_2\text{O}$. Secondary and tertiary amines do not give this smell, so the test specifically identifies primary amines.

1 mark Which reaction converts an amide to a primary amine with one fewer carbon atom?

MODEL ANSWER

Hoffmann bromamide degradation (with $\text{Br}_2 + \text{NaOH}$).

Ch 14 Biomolecules

MEDIUM

💡 Short and very scoring — pure memory. Glucose reactions, peptide bond, denaturation, DNA vs RNA are common 2–3 mark items.

IMPORTANT TOPICS (EXPLAINED)

Carbohydrates

Classified as **monosaccharides** (glucose, fructose), **disaccharides** (sucrose — non-reducing; maltose, lactose — reducing), **polysaccharides** (starch, cellulose, glycogen). Glucose is an aldohexose; it does not react with Schiff's reagent or NaHSO_3 (suggesting a cyclic hemiacetal form). Anomers (α/β), glycosidic linkage.

Proteins

Made of **α -amino acids** joined by **peptide (amide) bonds**. Amino acids exist as **zwitterions** at the isoelectric point. Structure levels: primary (sequence), secondary (α -helix, β -sheet via H-bonds), tertiary, quaternary. **Denaturation** destroys secondary/tertiary structure (heat, acid) — function lost, e.g. boiling an egg.

Enzymes & vitamins

Enzymes are biological catalysts (mostly proteins), highly specific. **Vitamins**: fat-soluble (A, D, E, K) and water-soluble (B group, C). Deficiency diseases (scurvy — C, rickets — D, night blindness — A).

Nucleic acids

DNA and **RNA** are polymers of nucleotides (base + sugar + phosphate). DNA: deoxyribose, bases A-T-G-C, double helix; carries genetic information. RNA: ribose, bases A-U-G-C, usually single-stranded; involved in protein synthesis.

MUST-KNOW FORMULAS & REACTIONS

Peptide bond: —CO—NH— (formed between —COOH of one amino acid and —NH_2 of the next, losing H_2O)

DNA bases pair: Adenine–Thymine, Guanine–Cytosine; RNA has Uracil instead of Thymine

Glucose: aldohexose, $\text{CHO—(CHOH)}_4\text{—CH}_2\text{OH}$; cyclic pyranose hemiacetal (α/β anomers)

HOW QUESTIONS COME FROM THIS CHAPTER

- ▶ 2-mark: differences between DNA and RNA.
- ▶ 2-mark: what is a peptide bond / zwitterion / denaturation of proteins?
- ▶ 2-mark: distinguish reducing and non-reducing sugars / why is sucrose non-reducing?
- ▶ 1-mark: which vitamin deficiency causes scurvy? (vitamin C).

PRACTICE QUESTIONS WITH MODEL ANSWERS

2 marks Give two differences between DNA and RNA.

MODEL ANSWER

- DNA contains the sugar **deoxyribose** and the base thymine; RNA contains **ribose** and uracil (instead of thymine).
- DNA is usually **double-stranded** (double helix) and stores genetic information; RNA is usually **single-stranded** and takes part in protein synthesis.

2 marks What is denaturation of a protein? Give one example.

MODEL ANSWER

Denaturation is the loss of the native secondary and tertiary structure of a protein (by heat, change in pH, or chemicals) while the primary structure (sequence) stays intact. The protein loses its biological activity. **Example:** coagulation of egg white (albumin) on boiling, or curdling of milk.

2 marks Why is sucrose called a non-reducing sugar?

MODEL ANSWER

In sucrose the reducing groups of glucose (its aldehyde/anomeric –OH) and fructose (its keto/anomeric –OH) are both involved in the **glycosidic linkage**. Because no free reducing (anomeric) group is available, sucrose cannot reduce Tollens' or Fehling's reagent, so it is non-reducing.

1 mark Deficiency of which vitamin causes scurvy?

MODEL ANSWER

Vitamin C (ascorbic acid).

Ch 15 Polymers

CORE

💡 Easy, descriptive marks. Classification, addition vs condensation, and naming the monomers of common polymers are frequent.

IMPORTANT TOPICS (EXPLAINED)

Classification of polymers

By **source**: natural (rubber, cellulose), semi-synthetic (cellulose acetate), synthetic (nylon). By **structure**: linear, branched, cross-linked. By **mode of polymerisation**: addition vs condensation. By **molecular forces**: elastomers, fibres, thermoplastics, thermosetting.

Addition vs condensation polymerisation

Addition (chain growth): unsaturated monomers add repeatedly with no loss of small molecules — polythene, PVC, polystyrene, Teflon (via free-radical/ionic mechanism).

Condensation (step growth): bi-functional monomers combine with elimination of a small molecule (H_2O , HCl) — nylon-6,6, terylene, bakelite.

Important polymers & their monomers

- **Polythene** — ethene.
- **PVC** — vinyl chloride.
- **Teflon** — tetrafluoroethene.
- **Nylon-6,6** — hexamethylenediamine + adipic acid.
- **Terylene (Dacron)** — ethylene glycol + terephthalic acid.
- **Bakelite** — phenol + formaldehyde.
- **Natural rubber** — isoprene; vulcanisation (with S) hardens it.

Copolymers & biodegradable polymers

Copolymers form from two different monomers (e.g. Buna-S = butadiene + styrene).

Biodegradable polymers decompose naturally — e.g. PHBV, Nylon-2-nylon-6 — reducing pollution.

MUST-KNOW FORMULAS & REACTIONS

Polythene: $n \text{ CH}_2=\text{CH}_2 \rightarrow -(\text{CH}_2-\text{CH}_2)-n$ (addition)

Nylon-6,6: hexamethylenediamine + adipic acid $\rightarrow$ polyamide + H_2O (condensation)

Terylene: ethylene glycol + terephthalic acid $\rightarrow$ polyester + H_2O

Bakelite: phenol + HCHO → cross-linked phenol-formaldehyde resin

HOW QUESTIONS COME FROM THIS CHAPTER

- ▶ 2-mark: difference between addition and condensation polymers with examples.
- ▶ 2-mark: name the monomers of nylon-6,6 / terylene / bakelite / Buna-S.
- ▶ 1-mark: what is vulcanisation? / give one biodegradable polymer.
- ▶ 2-mark: classify polymers based on molecular forces (elastomer, fibre, thermoplastic, thermoset).

PRACTICE QUESTIONS WITH MODEL ANSWERS

2 marks Distinguish between addition and condensation polymerisation with one example each.

MODEL ANSWER

Addition polymerisation: unsaturated monomers join repeatedly without the loss of any small molecule; the polymer has the same empirical formula as the monomer. e.g. polythene from ethene.

Condensation polymerisation: two bi-functional monomers combine with the elimination of a small molecule such as water. e.g. nylon-6,6 from hexamethylenediamine and adipic acid.

2 marks Name the monomers used to make (a) nylon-6,6 (b) terylene (c) bakelite.

MODEL ANSWER

(a) Nylon-6,6 — **hexamethylenediamine and adipic acid.**

(b) Terylene — **ethylene glycol and terephthalic acid.**

(c) Bakelite — **phenol and formaldehyde.**

1 mark What is vulcanisation of rubber?

MODEL ANSWER

Heating natural rubber with **sulphur**, which forms cross-links between polymer chains, making the rubber harder, stronger and more elastic.

💡 Pure memory, easy marks. Classes of drugs, soaps vs detergents and food additives are quick to revise and score.

IMPORTANT TOPICS (EXPLAINED)

Drugs & drug–target interaction

Drugs interact with biological targets (enzymes, receptors). **Enzyme inhibitors** block active sites; **receptor antagonists** block the natural messenger. Chemotherapy uses chemicals against disease.

Classes of medicines

- **Antacids** (reduce stomach acid — ranitidine, $\text{Mg}(\text{OH})_2$).
- **Antihistamines** (allergy — relieve histamine effects).
- **Tranquilizers / antidepressants** (stress, anxiety).
- **Analgesics** (pain relief — aspirin; non-narcotic vs narcotic like morphine).
- **Antimicrobials / antibiotics** (penicillin) — broad/narrow spectrum, bactericidal/bacteriostatic.
- **Antiseptics** (applied to living tissue — Dettol) vs **disinfectants** (non-living surfaces — phenol).

Chemicals in food

Artificial sweeteners (saccharin, aspartame), **food preservatives** (sodium benzoate), antioxidants (BHT, BHA).

Cleansing agents — soaps vs detergents

Soaps are sodium/potassium salts of long-chain fatty acids; they don't work in hard water (form insoluble scum with $\text{Ca}^{2+}/\text{Mg}^{2+}$). **Synthetic detergents** work even in hard water; they may be anionic, cationic or non-ionic. Cleansing action: the hydrophobic tail dissolves grease, the hydrophilic head faces water, forming a **micelle** that is washed away.

MUST-KNOW FORMULAS & REACTIONS

Antiseptic vs disinfectant: antiseptic → living tissue (Dettol, dilute phenol); disinfectant → non-living surfaces (conc. phenol)

Soap: RCOO^-Na^+ (fails in hard water) ; Detergent: e.g. sodium lauryl sulphate (works in hard water)

Cleansing action: micelle – hydrophobic tail into grease, hydrophilic head into water

HOW QUESTIONS COME FROM THIS CHAPTER

► 2-mark: difference between antiseptics and disinfectants with examples.

► 2-mark: difference between soaps and detergents / why soaps fail in hard water.

► 2-mark: explain the cleansing action of soap (micelle formation).

► 1-mark: give one example of an antacid / artificial sweetener / antibiotic.

PRACTICE QUESTIONS WITH MODEL ANSWERS

2 marks Differentiate between antiseptics and disinfectants with one example each.

MODEL ANSWER

Antiseptics are applied to living tissues (skin, wounds) to kill or prevent the growth of microorganisms, e.g. Dettol, dilute phenol, tincture of iodine.

Disinfectants are applied to non-living surfaces (floors, drains, instruments), e.g. concentrated phenol, chlorine, SO_2 . The same substance can act as either, depending on concentration (e.g. phenol).

2 marks Why do soaps not work well in hard water, and how do detergents solve this?

MODEL ANSWER

Hard water contains Ca^{2+} and Mg^{2+} ions, which react with soap (sodium salts of fatty acids) to form insoluble **scum** (calcium/magnesium salts), wasting the soap and reducing lathering. **Synthetic detergents** are salts of sulphonic acids/sulphates whose calcium and magnesium salts are **soluble**, so they lather and clean even in hard water.

2 marks Explain the cleansing action of soap.

MODEL ANSWER

A soap molecule has a long hydrophobic (water-repelling) hydrocarbon tail and a hydrophilic (water-attracting) -COO^- head. In water, the tails dissolve into the grease/oil on the dirty surface while the heads point outward into the water, forming a spherical **micelle** that traps the grease at its centre. These micelles are washed away with water, removing the dirt.

1 mark Give one example of an antacid.

MODEL ANSWER

Ranitidine (or magnesium hydroxide / sodium bicarbonate).

 Exam pattern — how the 60 marks are split

- **Total questions: 31.** You don't answer all of them — there is internal choice. Read instructions on the day.
- **Q1–5 · 1 mark each** (objective / fill-in / match / one-word) — attempt any 4 → 4 marks.
- **Q6–15 · 2 marks each** (very short) — attempt any 8 → 16 marks.
- **Q16–26 · 3 marks each** (short) — attempt any 8 → 24 marks.
- **Q27–31 · 4 marks each** (long) — attempt any 4 → 16 marks.
- **Theory total = 60.** Internal/practical = 40. Grand total = 100.
- **You must score at least 18/60 (30%) in theory to pass.** There is **no negative marking**, so attempt everything.

 How to write answers and grab easy marks

- **Always draw the structure / equation.** In organic questions a correct balanced equation alone can fetch most of the marks even if your words are weak.
- **Balance every chemical equation** and write conditions over the arrow (catalyst, temperature, reagent). Examiners give a separate mark for conditions.
- **Numericals: write the formula first, substitute, then answer with units.** You get step marks even if the final number is wrong. Never skip the formula line.

- **Definitions: give the one-line definition + one example.** "Define + example" pattern wins full 2-mark questions.
- **Distinguishing tests: name the reagent + state what each compound does** (e.g. "aldehyde gives silver mirror with Tollens', ketone does not").
- **Use keywords.** Markers look for terms like "nucleophilic addition", "lanthanoid contraction", "colligative", "spectrochemical series". Sprinkle them in.
- **Attempt the choice question you know best.** With internal choice, pick the version where you can draw a clean diagram.
- **Time: ~2 min per mark.** Don't over-write a 1-mark answer. Bullet points are fine and faster to mark.

Sources used to build this (official syllabus + exam pattern):

[Kerala Plus Two Syllabus & mark distribution — Careers360](#) · [DHSE Kerala exam pattern & marking scheme — Careers360](#) · [Chapter-wise weightage — HSSLive](#) · [Plus Two Chemistry chapter notes — A Plus Topper](#)

Built as original revision material from the standard Kerala/NCERT Class 12 syllabus. Verify the live mark-scheme on dhsekerala.gov.in on exam day. Progress (ticked chapters, exam time) is saved in your browser only. Good luck — you've got this. 🍀